

Electrical Load Forecasting Utilizing an Explainable Artificial Intelligence (XAI) Tool on Norwegian Residential Buildings

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Abstract—Electrical load forecasting is an essential part of the smart grid to maintain a stable and reliable grid along with helping decisions for economic planning. With the integration of more renewable energy resources, especially solar photovoltaic (PV), and transitioning into a prosumer-based grid, electrical load forecasting is deemed to play a crucial role on both regional and household levels. However, most of the existing forecasting methods can be considered black-box models due to deep digitalization enablers, such as Deep Neural Networks (DNN), where human interpretation remains limited. Additionally, the black box character of many models limits insights and applicability. In order to mitigate this shortcoming, eXplainable Artificial Intelligence (XAI) is introduced as a measure to get transparency into the model's behavior and human interpretation. By utilizing XAI, experienced power market and system professionals can be integrated into developing the data-driven approach, even without knowing the data science domain. In this study, an electrical load forecasting model utilizing an XAI tool for a Norwegian residential building was developed and presented.

Index Terms—Electrical Load Forecasting, Explainable Artificial Intelligence (XAI), Residential Buildings

I. INTRODUCTION

With the recent improvements in living standards, the electricity demand has significantly increased over the last decade. The transition towards a sustainable grid system is also challenging for the operation and stability of existing electric power systems due to the high electricity demand. However, the demand for electricity is expected to only increase at least till 2050 [1]. Whereas in Norway, it is predicted that the yearly power consumption will increase by 23 TWh, excluding the electrification of the oil sector [2].

Additionally, with the green shift in the electricity sector, more volatile means of power production are introduced to the grid, such as solar (PV) and wind. This uncertainty due to the volatility of such sources coupled with the introduction of large loads as a result of increased electrification rate will result in greater stress for the existing grid infrastructure. Statnett, who is responsible for the operation and construction of the Norwegian power grid, has in their investment plan for 2021 an estimated 60 – 100 billion NOK in grid investments [3].

Residential households are accountable for 22% of all energy usage in Norway, which is mainly driven by electricity [4] and are especially susceptible to power peaks. As electricity carries the characteristic that it needs to be used as soon

as it is produced further establishes the need for good and understandable forecasting. AI-based solutions have been used in the electric grid for various use cases such as network operation and control, energy management, energy generation and load forecasting [5].

Such solutions have been critical in the modernization of the electrical grid and making it a “smart” grid. Advancements in Computer Intelligent (CI) models have also created complex forecasting tools using Artificial Neural Networks (ANN). However, a disadvantage of highly complex models, such as ANN, is that they are impossible for even the most competent data scientist to understand.

Over the last decade, XAI models have been increasingly popular as presented by Arrieta et al. [6]. XAI can increase interpretability, which can lead to further accuracy during the model development [7]. XAI has also seen steps in the energy sector, an article performed by Kruse et al. [5] explored the opportunities of incorporating XAI into frequency control of power systems. In another study, Zhang et al. [8] applied an XAI method called SHapley Additive exPlanations (SHAP) for emergency control of power systems. There has however been little work done in the field of energy forecasting, one study by Kuzlu et al. [9] explored different XAI models (SHAP, LIME, ELI5) on solar generation forecasting. Meanwhile, Grimaldo et al. in [10] used the combination of XAI with an interactive dashboard to allow the user to understand how the input parameters affect the results as well as provide different forecasting models.

This study focuses on the hour ahead electricity load forecasting using a combined method, i.e., Long Short Term Memory (LSTM) and the XAI tool (SHAP), to interpret the forecasting results. By utilizing the XAI tool, key factors that are affecting the forecast were identified to improve the accuracy of the model. The main contributions of this paper are (1) Developing an LSTM-based local electrical load forecasting model for Norwegian residential buildings, (2) Apply the SHAP XAI tool in order to have quantitative interpretation results and identify the key parameters within the model, (3) Investigate the impact of Numerical Weather Predictions (NWP) and other input features coupled with various XAI tools on the accuracy of the XAI-based forecasting results.

II. BACKGROUND

A. Recurrent Neural Networks (RNN)

Recurrent Neural Network (RNN) is a sub-field of deep learning which excels at processing sequence data. As a RNN is great at understanding series of data, these kind of networks are usually used for language translation and time series forecasting. The major difference of RNN compared to MultiLayer Perceptrons (MLP) is RNN's internal recurrence. As the name implies the hidden layers of a RNN has recurrent connections within each neuron. One way to formulate a RNN is as done by Goodfellow et al. [11], by the following equations,

$$\mathbf{a}^{(t)} = \mathbf{b} + \mathbf{W}\mathbf{h}^{(t-1)} + \mathbf{U}\mathbf{x}^{(t)}, \quad (1a)$$

$$\mathbf{h}^{(t)} = \Theta_1(\mathbf{a}^{(t)}), \quad (1b)$$

$$\mathbf{o}^{(t)} = \mathbf{c} + \mathbf{V}\mathbf{h}^{(t)}, \quad (1c)$$

$$\hat{\mathbf{y}}^{(t)} = \Theta_2(\mathbf{o}^{(t)}). \quad (1d)$$

Equation (1a) represents the forward propagation through the network, with $\mathbf{b}$ and $\mathbf{c}$ being the bias vectors. For each sequence of data, $\mathbf{x}^{(t)}$, a hidden state, $\mathbf{h}^{(t)}$ is calculated using equations (1a) and (1b) where Θ_1 usually denotes the $\tanh$ activation function. The model output $\hat{\mathbf{y}}^{(t)}$ is calculated using equations (1c) and (1d), where the matrices $\mathbf{U}$, $\mathbf{V}$ and $\mathbf{W}$ are the weights between input, hidden layer and output. After forward propagating through the system and the cost function is calculated the weights are updated using the back-propagation algorithm.

However, as RNNs are calculated recurrently, the back-propagation is applied similarly and thus the weights are updated in each time step that is being calculated. This does not only make back-propagation time-consuming but it also makes RNNs vulnerable to vanishing and exploding gradients. Calculating the gradients for long sequences using the chain rule over each time step can give either small or large value as each derivative is calculated with the prior time steps in back-propagation.

B. Long Short-Term Memory (LSTM)

LSTM is an advancement of RNN with the goal of reducing the vanishing gradient problem. Hochreiter et al. [12] proposed replacing the hidden units in the RNN with LSTM cells where each cell is equipped with three gates; forget, input and output. In addition to these three gates a cell state vector, $\mathbf{C}^{(t)}$, connects each LSTM cell time step, t . In the cell state, valuable information is contained, which is passed on to the next cell. The gates are responsible for determining what information that is important. During the forward pass the forget gate decides what information to remove from input, $\mathbf{x}^{(t)}$ and previous output, $\mathbf{h}^{(t-1)}$. This is mathematically formulated in

equation (2a), where $\mathbf{U}$ represents the input weight whilst $\mathbf{b}_f$ is the bias vector for the forget [13].

The Sigmoid function, σ is applied to ensure that the output is differentiable and between 0 and 1. The input gate decides what information is going to be kept in the cell state. This is combined with potential values from the previous output and current input and is formulated in equations (2b) - (2d). $\tilde{\mathbf{C}}$ is the candidate vector, containing the potential new information, which is added to the cell state in equation (2d). The candidate vector is calculated using the $\tanh$ function to keep the output between -1 and 1. Here, b_i and b_C represents bias for input and candidate vector. Furthermore, $\mathbf{W}_c$ denote the weight matrix for the cell state. The output of the LSTM cell is calculated using the updated cell state and is described by equations (2e) and (2f) where the output gate creates a vector, $\mathbf{o}$, which identifies what parts of the cell state that is supposed to be sent through [13].

$$\mathbf{f}^{(t)} = \sigma(\mathbf{U} \cdot [\mathbf{h}^{(t-1)}, \mathbf{x}^{(t)}] + \mathbf{b}_f), \quad (2a)$$

$$\mathbf{i}^{(t)} = \sigma(\mathbf{V} \cdot [\mathbf{h}^{(t-1)}, \mathbf{x}^{(t)}] + \mathbf{b}_i), \quad (2b)$$

$$\tilde{\mathbf{C}}^{(t)} = \tanh(\mathbf{W}_c \cdot [\mathbf{h}^{(t-1)}, \mathbf{x}^{(t)}] + \mathbf{b}_C), \quad (2c)$$

$$\mathbf{C}^{(t)} = \mathbf{f}^{(t)} * \mathbf{C}^{(t-1)} + \mathbf{i}^{(t)} * \tilde{\mathbf{C}}^{(t)}, \quad (2d)$$

$$\mathbf{o}^{(t)} = \sigma(\mathbf{W} \cdot [\mathbf{h}^{(t-1)}, \mathbf{x}^{(t)}] + \mathbf{b}_o), \quad (2e)$$

$$\mathbf{h}^{(t)} = \mathbf{o}^{(t)} * \tanh(\mathbf{C}^{(t)}). \quad (2f)$$

C. eXplainable Artificial Intelligence (XAI)

The application of XAI can be divided into model agnostic and model specific where model agnostic methods can be defined as methods, which are universal across different machine learning algorithms. Such methods are beneficial as it gives the opportunity to compare the output of multiple models [14] and thus can be considered an advantage over model specific which only can be used on specific algorithms, limiting when they can be used.

Important traits to a model agnostic method beyond being able to be used for different models is explanation, representation and flexibility, which entails different types of explanations based on the problem at hand [15]. XAI methods are also divided into local and global, describing individual predictions and average behavior, respectively [14]. In this research, SHAP has been chosen over other XAI methods such as LIME due to its ability for offering both local and global explanations. Additionally, SHAP library offers various visualization tools and plots, making understanding the model easier for domain experts.

The methodology of SHAP is based on game theory, where a ranking is performed on the different participants' influence in a game to determine what share of the return they must get. In SHAP, this is done by creating a combination of each

possible subset of participants and recalculating the outcome of the game. This gives an indication of how each group member contributed to the task and how different groups would fair. Lloyd S. Shapley proposed the following mathematical explanation for Shapley values,

$$\varphi_j(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus j)]. \quad (3)$$

The goal is to formulate this as a Machine Learning (ML) problem and observe how the different features impact the outcome of the model. From equation (3) the Shapley value of a feature j is calculated using the model f and a feature vector x [14]. However, due to model complexity, a simplification of the features into x' may be necessary. The first part of Equation (3) can be viewed as a weight, which incorporates how adding or subtracting different features impact the model based upon all features M . $[f_x(z') - f_x(z' \setminus j)]$ shows the contribution of a subset of features, i in z' . Since Shapley values are calculated on all subsets of features, the numbers of calculations become exponentially high.

SHAP breaks features down to $z' \subseteq \{0, 1\}^M$ indicating that a feature is either viewed as active or asleep. One formal way of expressing SHAP defined by Lundberg et al. [16] is,

$$g(z') = \varphi_0 + \sum_i^M \varphi_i z'_i. \quad (4)$$

Where g signifies an interpretable model in the set of potential global models G . There are multiple ways of calculating SHAP values; however, in this research, the Deep SHAP technique was utilized [16] which takes the advantage of the knowledge of DNN. This is done by combining SHAP values for the whole network with calculated values for smaller parts of the network which is similar to back-propagation performed by neural networks [16].

III. METHODOLOGY

The flowchart in Figure 1 visualizes the proposed methodology for this work. The raw data includes historical net load data from a residential home located near Oslo as well as NWP which is provided daily at 18:00 by The Meteorological Institute of Norway [17]. As can be seen, the first step is data inspection for missing values and visualization to establish first insights about the dataset such as historical patterns. Later on, the feature engineering procedure based on Principle Component Analysis (PCA), as well as data standardization, was performed.

After the preprocessing and visualization of data for initial analysis, test and train dataset splits were performed for the AI model development segment. The training and testing data is used to train and develop multiple models such as CNN, RNN, and LSTM. The models are trained to utilize historical load data as well as NWP data to get insight into how ambient temperature, pressure, and wind effects the model performance. AI model development is done using the Python

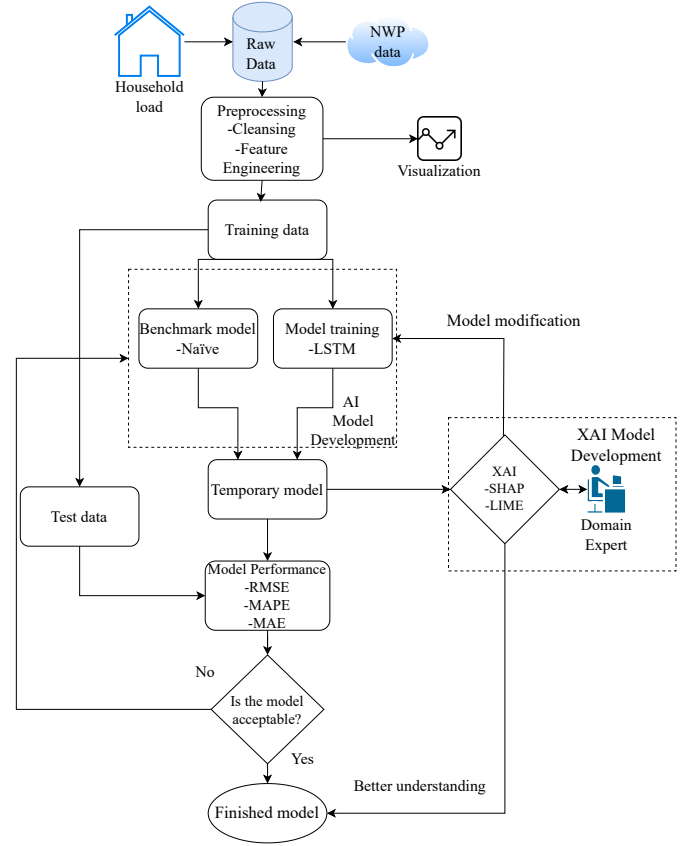


Fig. 1. The proposed methodology for the XAI-based electrical load forecasting and XAI utilization.

libraries Sklearn and Tensorflow. Algorithm hyperparameters such as learning rate, activation functions, and epochs are optimized using the neural net library Keras in Tensorflow. After observing the outcome of the models, LSTM was picked for application of XAI method for the scope of this paper. In addition to the LSTM model a naïve forecast is performed as a baseline to compare the the proposed model.

In order to evaluate the developed models' performances, XAI tool SHapley Additive exPlanations (SHAP) is used [9], [16] for providing more insight and explanation for the model developers and other domain experts with less expertise in data science. The traditional forecast performance metrics, such as Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE) and Mean Absolute Error (MAE), are used to determine forecast accuracy. The results from the XAI tools and the performance measures serve as indicators to further enhance the model by tuning parameters and feature engineering. The output of the XAI model serves as an indicator for the domain experts to further improve the model by understanding which input parameter the model relies more heavily.

A. Data Collection and Preprocessing

The unprocessed weather data was filtered for missing values as well as erroneous measurements, which were re-

placed by interpolation using the neighbouring data points. NWP variables, which were utilized in this research were: temperature, humidity, wind, pressure, and cloud coverage. Wind values are retrieved as vectors with an X and Y direction. Instead of wind direction which was viewed as arbitrary, wind magnitude was included in the final prepared dataset.

The electrical load data is retrieved from Elhub, the central hub for smart meter values in Norway [18]. 3 years of hourly data (25368 data points) was collected and divided into training (72%), validation (8%), and test (20%) set. Afterwards, the training dataset was normalized due to different scales within the dataset. Each prediction uses the previous 24 data points as input features, giving the input array in a form of (length of training data, number of previous time steps used, number of input features). Additionally, in order to determine their effects on the model, several one-hot encoded parameters were added as extra features such as seasonality, months, weekdays and hours.

1) *AI Model development*: First step of the model development was the preparation of train and test datasets. The utilized dataset consists of 3 years of data with hourly resolution, hence in order to allow model to train on most of the data, train and test splits were done by 80-20% respectively. The independent variables (X) within the model are: Historical hourly load values for the previous 24 hours, 24 day ahead NWP data with 1 hour resolution, and 7 one hot encoded day of the week and 24 one hot encoded hour of the day parameters. Meanwhile, the target parameter (y) is the residential load value $P(t)$. The input data was transformed into tensors by shape (number of samples, number of lags, number of features), whereas the y tensor has the shape of (number of samples, number of target variables).

The utilized model consists of two LSTM layers with the addition of a dense layer after the second layer in order to reshape the data into a single output. Additionally, a dropout layer is added between each LSTM layer in order to mitigate overfitting. In addition to dropout layers early stopping is implemented to further stop overfitting. Meanwhile, to counteract exploding gradients, a gradient clipping was implemented into the model.

Hyperparameter tuning was done via the RandomSearch function within KerasTuner Python library [19]. Each model was tuned using 75 trials where each trial was ran for 10 epochs where the total optimization time was approximately 7 hours. The final parameter results after hyperparameter tuning for the developed LSTM model was; Batch size: 32, Dropout: 0.2, Layer-1: 116 units, Layer-2: 118 units, Optimizer: Adam, clipvalue: 0.5, learning rate: 0.005, Training time: 64.89 seconds, Epochs: 4.

2) *XAI Model development*: As mentioned, SHAP can be utilized for providing global or local explanations as well as different ways of visualizing the calculated values [16]. The Shapley values were calculated by utilizing the DeepExplainer using 1000 random samples from the training data as a foundation [20].

The entire test set was utilized for the SHAP via the

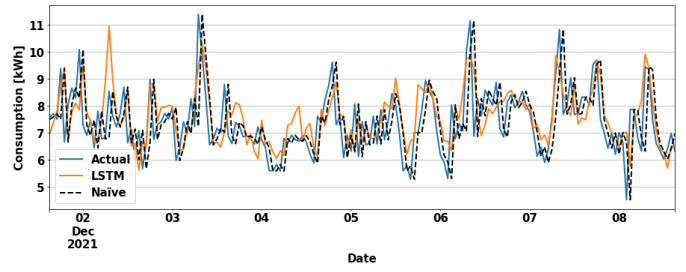


Fig. 2. Electrical load forecast during a week in December 2021, Comparison of actual load and LSTM forecasting.

estimates from the random samples. The SHAP values were returned as a tensor with the form off (# samples, # lags, # features). As this model operates using 24 lags, each output resulted in $24 \times \#features$ for each sample. To observe the impact of all features, the SHAP values were reshaped into (# lags $\times$ # samples, # features).

Due to the high number of information displayed, heatmap, as well as summary plots, were utilized for visualizing global SHAP explanation values. On the other hand, a waterfall plot was used for displaying the local SHAP explanation values. In addition to this, a decision plot was used for observing the local explanations over a short time frame.

IV. RESULTS

The final model utilized Load, NWP, and one-hot encoded hours, and days as input features. The LSTM model's performance is depicted in Table I and MAPE, RMSE, MAE are shown for each month as well as the performance for the entire test set. The naïve forecast is similarly presented in Table II. Figure 2 portrays a week of predictions compared to the actual load, which is visualized in blue. Additionally, the naïve forecast is presented as a dashed line. From Figure 2, it can be observed that the model output is able to follow the actual load. However, it struggles with the sharp sudden load changes. The calculated MAPE for this time period is 0.11292, and RMSE is 1.00900, making it more accurate compared to the total performance. From Table I, it is worth noting that MAPE performance increases during the winter, while RMSE is fractionally worse during the winter months compared to summer.

Figure 3 illustrates feature impact on all instances. Shades of blue indicate the instance to have lower value than the expected

TABLE I
PERFORMANCE OF EACH MONTH AND THE ENTIRE PERIOD FOR LSTM,
CALCULATED IN MAPE, RMSE AND MAE.

Month	MAPE	RMSE	MAE
July	0.3071	0.8333	0.5938
August	0.2878	0.9767	0.7326
September	0.2669	0.9100	0.6873
October	0.2216	0.9168	0.6853
November	0.2166	1.053	0.7702
December	0.2147	1.103	0.8503
Total	0.2622	0.9748	0.7251

TABLE II
PERFORMANCE OF EACH MONTH AND THE ENTIRE PERIOD FOR NAÏVE,
CALCULATED IN MAPE, RMSE AND MAE.

Month	MAPE	RMSE	MAE
July	0.3135	0.9434	0.6382
August	0.3230	1.182	0.8456
September	0.2947	1.057	0.7708
October	0.2505	1.099	0.8060
November	0.2173	1.227	0.8706
December	0.1719	1.115	0.7948
Total	0.2721	1.122	0.7973

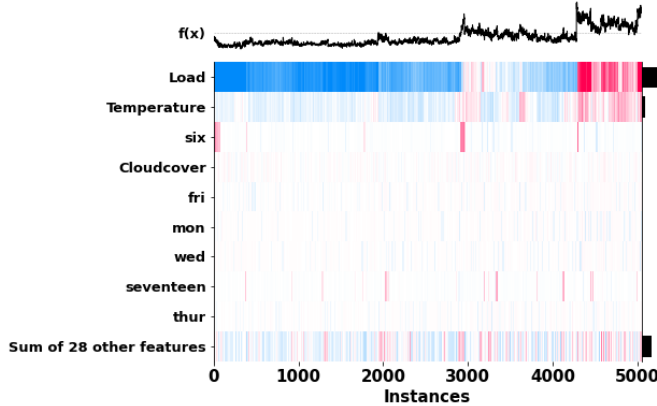


Fig. 3. Heatmap of the entire test set.

value of the entire model, whereas shades of red signify the opposite. The importance of each feature is ranked on the Y-axis with the most important feature being at the top. It is worth noting that the three most overall important features are: Load, temperature, and six, which denotes the time 06:00. The time importance can be due to morning peak demand which can result in the shower, food preparation, etc. It is also observable that the hour 17:00 (seventeen in Figure 3) has high importance too, which corresponds to the typical time Norwegians arrive home from work.

Figure 4 depicts a summary plot showing the importance of the different features on the y-axis. On the x-axis each explanation is visualized as a colored dot. As with other plots, the shade of the color signify the relative value of the output. The impact of each value, also known as SHAP value is shown along the x-axis, with explanations increasing the output having a high positive value and explanations decreasing the output having a high negative value. It is observed that high historical load values produce similar outputs with high importance and the load is the most important feature. For temperature, it is seen that low temperatures (blue dots) increase the model output coinciding with logic that colder temperatures increase electric load.

The decision plot in Figure 5 shows how the features impact the model for each prediction on the 8th of December 2021. 07:00 and 09:00 are highlighted with dashed lines. It is

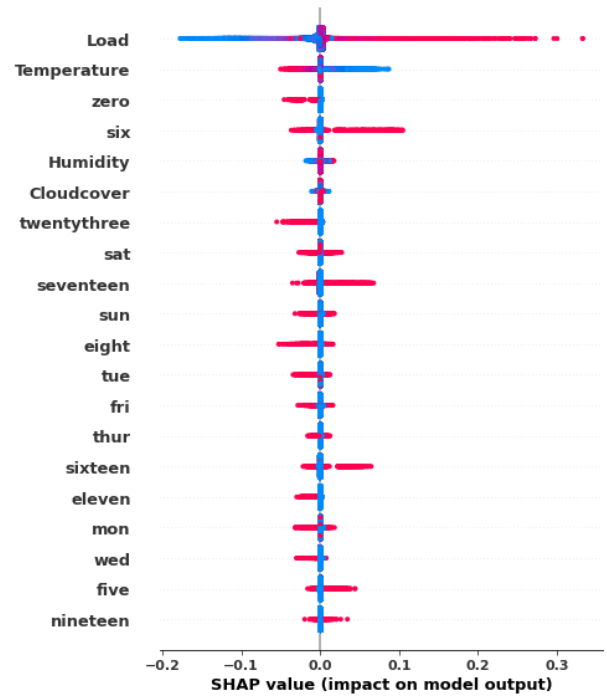


Fig. 4. Average influence of each feature on the entire test set.

observed that load and temperature are important features for all predictions that day, which coincides well with Figure 3. Furthermore, for 07:00 one can see that the one-hot encoded hour for 06:00 is heavily emphasized.

Figure 5 shows feature influence for each forecasted hour on 2021-12-08, while Figure 6 gives a local explanation for

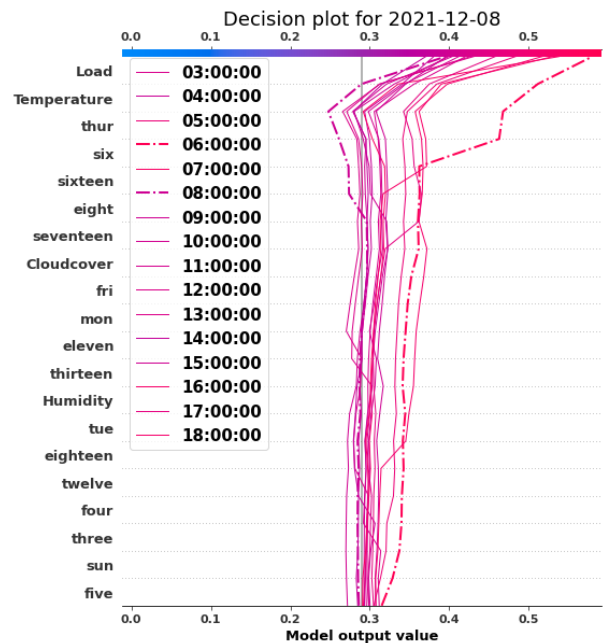


Fig. 5. Feature influence for each forecasted hour on 2021-12-08.

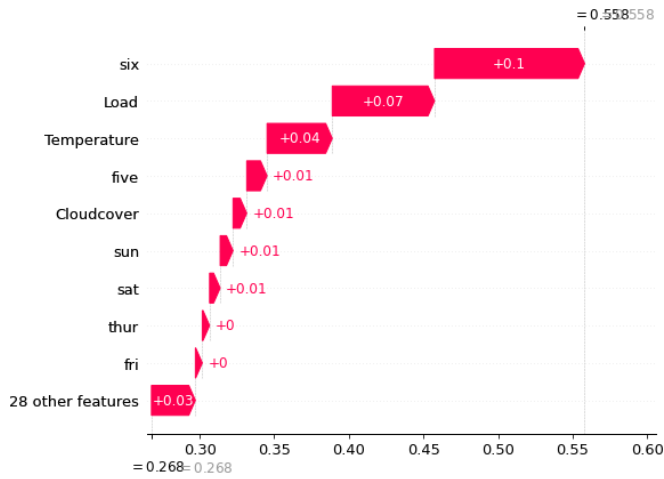


Fig. 6. Feature influence at 07:00 on 2021-12-08.

07:00 on the same day. This plot clearly depicts the three most important features: the previous hour, load, and temperature values. The other features are understood to have little to no impact on this forecast.

V. DISCUSSION AND CONCLUSION

From the SHAP explanations, it is clear that some features bring more importance than others. First of all, historical load values have the highest significance, as it closely represents the target variable. Furthermore, from Figure 3 it can be observed how the model identifies the impact of temperature in its forecasts. As the test set spans from June to December the season change can be seen by how temperature and load go from decreasing the predicted load (blue) to increasing the predicted load (red). This is understood as it goes from summer to winter and the temperature decreases it will be needed for more heating. This relation shows how the model has indirectly learned the seasonality of Norwegian weather patterns. During model development, a version, which included seasonality as the one-hot encoded feature was tested. However, it has shown no direct improvements to the model. This could be because the model had learned a more fluent way of identifying the seasons instead of a more absolute way based on the date. In Table I, the winter months are observed to outperform summer in regards to MAPE.

Comparing the naïve results in Table II to Table I it is observed that the naïve forecast performs surprisingly well, however is outperformed by the LSTM model in all statistics except for the MAPE in December. Why both models perform better during the winter is hard to pinpoint. However, a higher average load as well as more days spent away from home due to holidays could be a contributing factor.

It can also be observed that the timestamps with expected high power demand, such as early morning and afternoon had a high impact on the model. In Figure 4 it is seen that 06:00 and 17:00 impact the model by increasing the expected output. Furthermore, hours with expected low load, such as hours

close to midnight, are given a negative SHAP value, meaning that they decrease the model output. Meanwhile according to the plots, the wind speed and pressure from NWP data have little importance on the accuracy of the model. Hence it might be possible to drop those parameters and re-train the model in order to observe any accuracy improvements.

In this paper, an LSTM-based local residential load forecasting algorithm was used in combination with XAI tool SHAP in order to provide more insight into the black-box nature of such ANNs. The results of this study reveal the most important parameters that can be given more consideration during the forecastings as well as highlights the parameters that can be omitted in order to further increase the model accuracy.

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